

Task 2.2 — Perimeter & Access Control Design

Fencing & Security App

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Preliminary engineering concept
Perimeter · access · movement · validation



01 / PURPOSE & CONTEXT

1 From zoning to observable access control

This week's work translates the previous security zoning and risk/asset analysis into a preliminary perimeter and access-control concept.

The design question

The concept defines the farm boundary, organizes entry and exit points, links access levels to zone sensitivity, and makes physical protection observable.

Continuity

Task 1.3 identified risks and assets. Task 2.2 turns that analysis into visible boundaries, controlled points, and movement rules.

Five outputs for Task 2.2

2.2.1 Perimeter & boundary · 2.2.2 Entry & exit · 2.2.3 Access levels · 2.2.4 Physical protection · 2.2.5 Flow & mock-up

What is designed A traceable relationship between zone sensitivity, access control, physical protection, and movement.

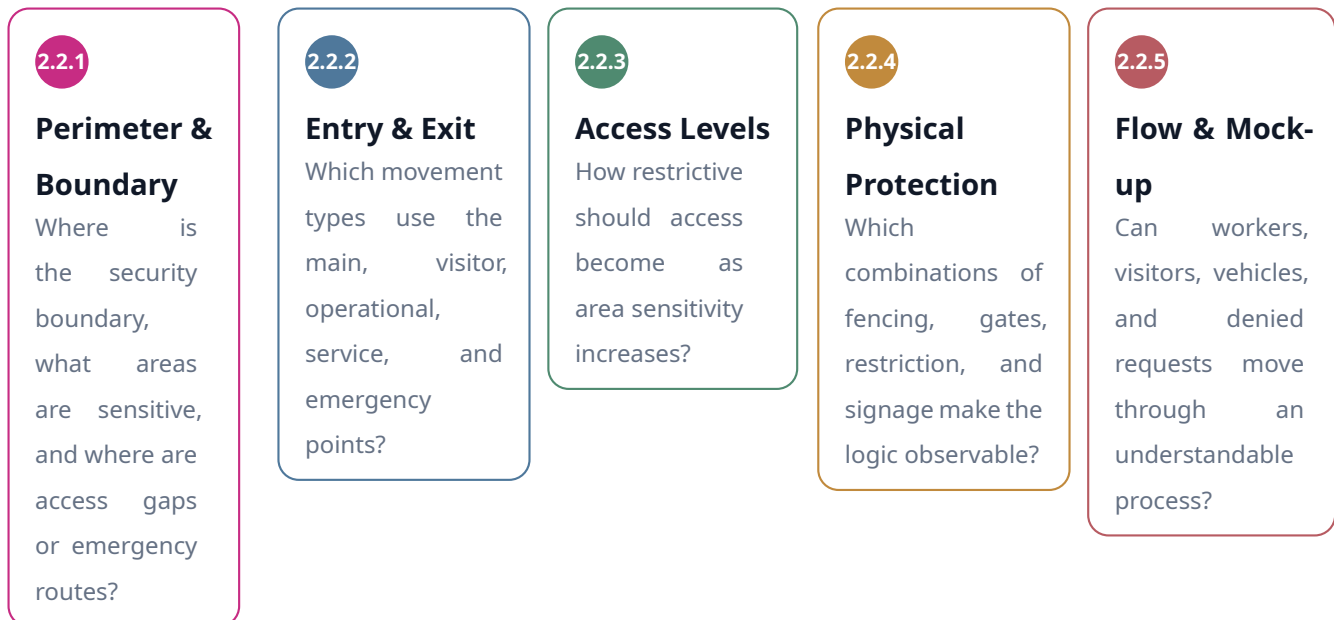
What is not claimed No exact site measurements, coordinates, fence dimensions, or construction specifications have been established.

Status Preliminary concept — to be validated on site against terrain, climate, routes, operations, maintenance, and emergency needs.

02 / DESIGN LOGIC

2 One connected security logic

The Miro-board hierarchy is retained as a progressive design chain: define the boundary, organize access points, assign levels, select physical responses, then test the idea through flows and a conceptual mock-up.



PERIMETER → ENTRY / EXIT → ACCESS LEVEL → PROTECTION → OBSERVABLE FLOW

Threat / requirement

Sensitive area; uncontrolled entry; operational traffic; emergency response; changing weather and local conditions.

Security objective → design response

Define the boundary; guide movement to visible points; restrict sensitive zones; preserve emergency access; support maintenance and traceability.

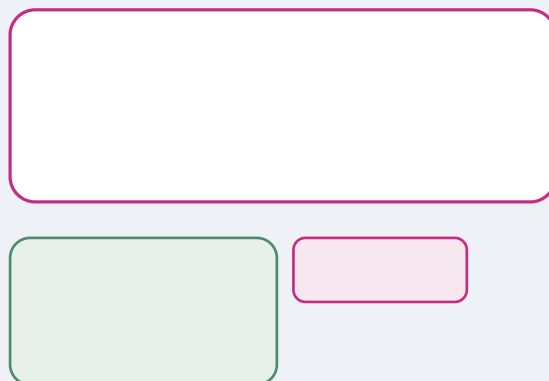
Design principle: each measure is proposed for a clear function, not as a guarantee of security.

3 Boundary first: make the security edge observable

Boundary components

- **Farm Boundary** Defines the outer security edge and the relationship to surrounding land.
- **Sensitive Areas** Identifies assets or activities requiring stronger control.
- **Access Gaps** Highlights weak, informal, or unobserved approach routes.
- **Emergency Access** Keeps response routes usable while avoiding uncontrolled everyday access.

CONCEPTUAL BOUNDARY



Sensitive zone + controlled point inside a defined edge

Engineering interpretation

The boundary is not only a fence line. It is a coordinated system of identification, controlled points, operational routes, and exceptions for emergency movement. The site must be surveyed before exact alignments, materials, dimensions, or gaps are defined.

Source support: USDA NRCS Access Control Code 472 [1]; FAO farm-structures reference [2].

4 Separate movement types without breaking farm operations

The entry system distinguishes movement types so that normal operations, visitors, services, and emergency response do not all rely on one undifferentiated route.

Main entrance

Primary arrival point for people and vehicles.
Provides a recognizable security boundary and an observable control location.

Visitor access

Directed to registration or verification before entering an authorized area. Visitor circulation remains legible and limited.

Staff / operational access

Supports routine worker movement through verification and assignment to an operational zone.

Service / emergency access

Supports deliveries, maintenance, emergency response, and other exceptional movements. Requires clear authorization and operational availability.

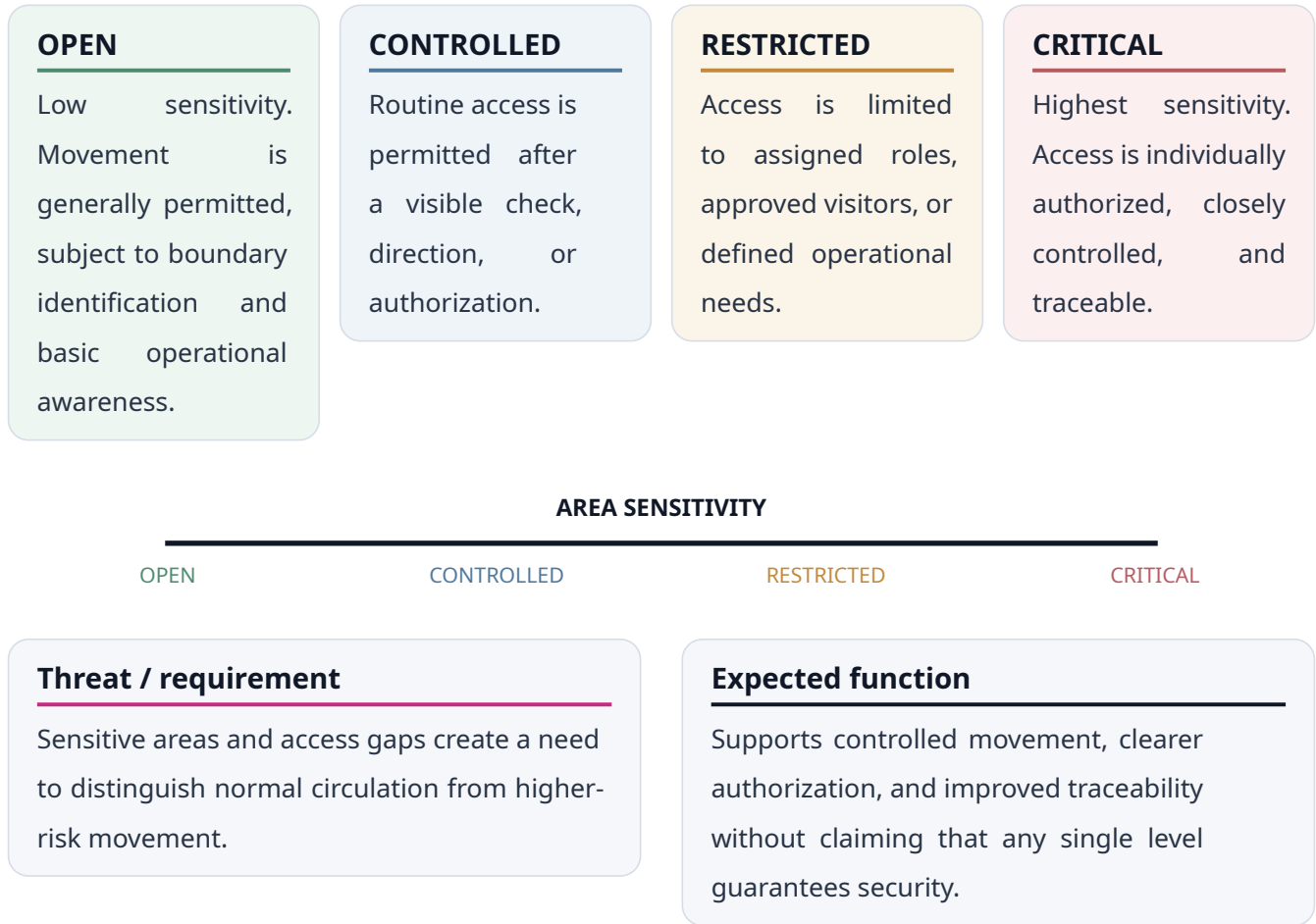
MAIN ENTRANCE**VISITOR****STAFF / OPS****SERVICE / EMERGENCY**

All points connect back to the defined boundary and the applicable access level.

Preliminary concept — exact gate locations and dimensions require site validation.

5 Restriction increases with area sensitivity

Four conceptual levels turn the zoning logic into an observable access rule. The levels are functional categories, not finalized technical specifications.



Source support: access control for sensitive areas [1]; flexible, scalable security planning [3].

6 Physical protection makes the concept visible

Four complementary responses

- 01 Fencing** Marks and supports the defined boundary.
- 02 Gates** Concentrate movement into observable entry and exit points.
- 03 Access restriction** Matches the control intensity to the assigned level.
- 04 Signage / identification** Makes the boundary and expected behavior understandable.

Relationship at a control point

FENCE / BOUNDARY

SENSITIVE AREA

GATE / CONTROL

Signage makes the boundary and expected behavior legible.

Preliminary design concept

The final fence type, gate hardware, dimensions, materials, and exact placements are intentionally not specified here. They must be selected after site validation of terrain, climate/weather exposure, operational routes, maintenance capability, emergency access, local conditions, and security requirements.

PRELIMINARY CONCEPT — TO BE VALIDATED ON SITE

7 Four flows, one traceable rule

WORKER

Staff / Operational Access → Verification →

Assigned Zone → Exit

Routine movement is tied to an operational role and destination.

VISITOR

Visitor Entry → Registration / Verification →

Authorized Area → Exit

Visitor circulation is bounded by authorization and an understandable route.

VEHICLE

Service Access → Verification → Operational

Area → Exit

Service and delivery movement remains compatible with farm operations.

UNAUTHORIZED ACCESS

Access Request → Denied → Stop / Redirect →

Incident Handling if required

The response is a managed handling path, not an unsupported promise of prevention.

VISIBLE POINT → VERIFY → AUTHORIZE / DENY → GUIDE MOVEMENT → EXIT / RESPOND

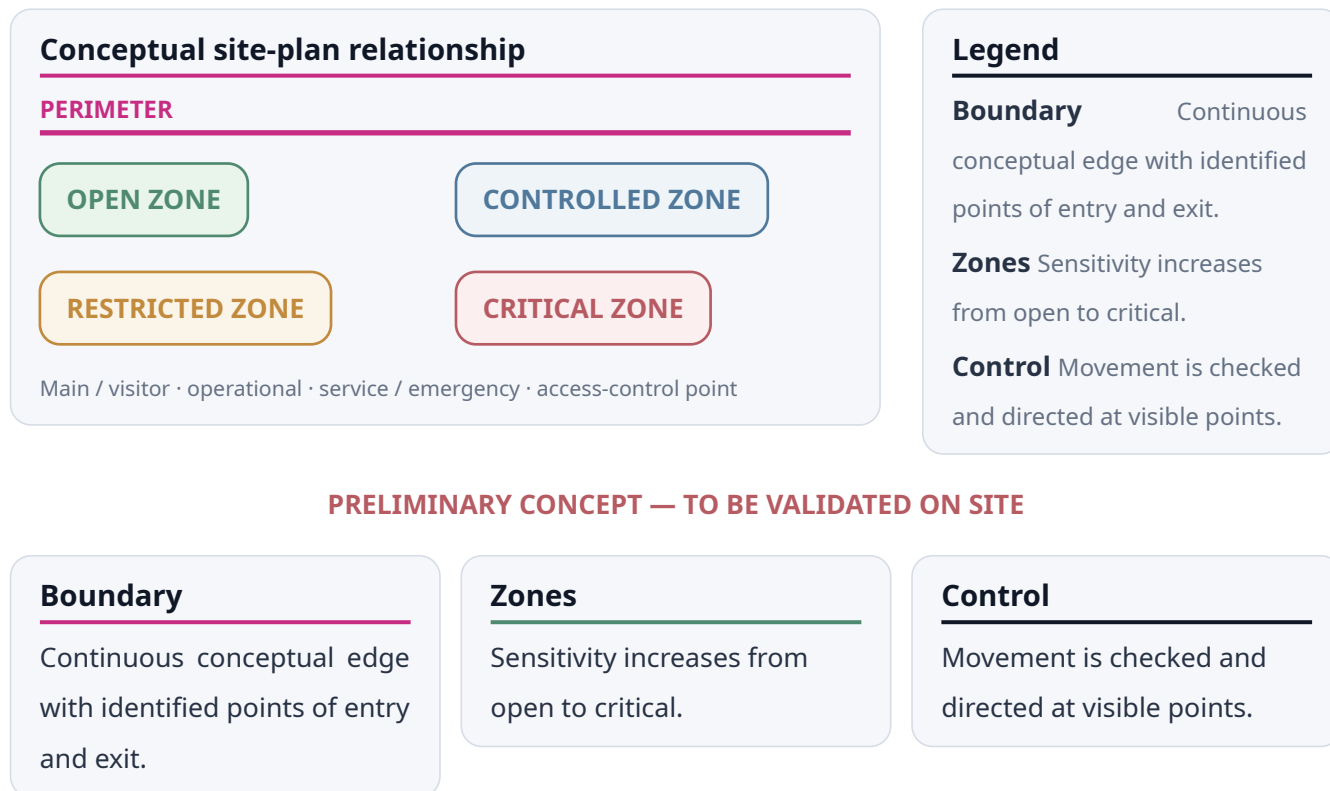
Why this matters

The flow is the operational bridge between the security zone and the physical site. It supports controlled movement, makes exceptions explicit, and gives the mock-up something observable to represent.

08 / CONCEPTUAL MOCK-UP

8 A simplified farm security model

This is a conceptual site-plan-style mock-up. It communicates relationships between boundaries, access points, zones, and a control point; it is not an actual geographic map.



9 What still needs validation

The concept is intentionally traceable but not final. The next engineering decisions depend on the actual site, the operating model, and the maintainability of the proposed controls.

Site and environment

Actual site layout; terrain; climate and weather exposure; local natural conditions; drainage and seasonal changes.

Operations

and movement

Operational routes; worker circulation; visitor handling; vehicle/service access; emergency movement and response.

Security and upkeep

Security requirements; access-level assignment; signage; maintenance capability; practical usability and inspection.

VALIDATION SEQUENCE

OBSERVE**TEST FLOWS****SELECT MEASURES****DOCUMENT**

Professional boundary

This report proposes a security and engineering concept. It does not claim a completed construction design, compliance certification, or guaranteed prevention of unauthorized access.

Site validation is a design requirement, not an optional afterthought.

10 / EVIDENCE & REFERENCES

10 Design principle → source → what it supports

The sources below are used for specific design principles only. They do not establish the farm's final layout or technical specifications.

DESIGN PRINCIPLE	SOURCE	WHAT THE SOURCE SUPPORTS
Access control can exclude people, vehicles, equipment, or animals from an area.	1 • USDA NRCS	The definition and applicability of access control to sensitive areas; use of signs, gates, and barriers as regulating measures.
Placement and operation must be described and maintained for the intended purpose.	1 • USDA NRCS	A preliminary concept should remain explicit about what is known, what must be selected, and what requires installation/maintenance validation.
Emergency contingencies must remain part of the access concept.	1 • USDA NRCS	Consideration of emergency-level contingencies such as wildfire, hurricane, drought, or flood, supporting the service/emergency access path.
Farm-structure decisions are site-dependent engineering decisions.	2 • FAO	The relevance of surveying, documentation, building maintenance, climate/environmental control, and fencing in rural farm structures.
Physical-security planning should be flexible, scalable, and resilience-oriented.	3 • CISA	A risk-informed planning posture using protective measures suited to varied physical environments and threat types.

References

[1] USDA NRCS, [Conservation Practice Standard: Access Control \(Code 472\)](#), July 2026.

[2] FAO/SIDA, [Farm structures in tropical climates](#), Food and Agriculture Organization of the United Nations, 1988.

[3] CISA, [Physical Security](#), U.S. Cybersecurity and Infrastructure Security Agency, accessed 20 August 2026.

11 Task 2.2 establishes a traceable access concept

The work connects previous security zoning to a preliminary, observable perimeter and access-control model for a farm environment.

What was designed

A boundary definition; movement-specific entry and exit points; four conceptual access levels; complementary physical protection; and a preliminary flow/mock-up.

What remains open

Exact site geometry, dimensions, materials, hardware, and final control arrangements remain to be validated against terrain, climate, operations, emergency needs, maintenance, and local conditions.

SECURITY ZONING



PHYSICAL ACCESS



CONTROLLED MOVEMENT

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PRELIMINARY ENGINEERING CONCEPT — TO BE VALIDATED ON SITE